CDSD block 2
Cyril Braguy

Speed dating with TINDER

Exploratory Data Analysis

Github project : cyrilbraguy/Bl2_Tinder

Pitch and GOAL

Pitch

- Tinder, the dating application, is facing declining matches and wants to understand **what makes people interested in each other.**
- A speed dating experiment has then been launched :

  four-minute dates in 21 waves plus questionnaire data teach us about decisions on second dates and ratings on six attributes, on demographics, habits, and lifestyle of people and speeddating partner.

Goal

- Use the dataset, through descriptive statistics and visualizations to identify what drives people to accept a second date.
- Explore questions such as attractiveness versus its real impact, shared interests versus shared background, self-perception accuracy, and date order. Deliver a notebook with statistics, charts, and interpretations.

Plan of analysis

- Load and clean data
- EDA of speed daters
- EDA of waves
- Analysis
- Conclusion

1 Load data

8378 lines 195 columns, 21 waves

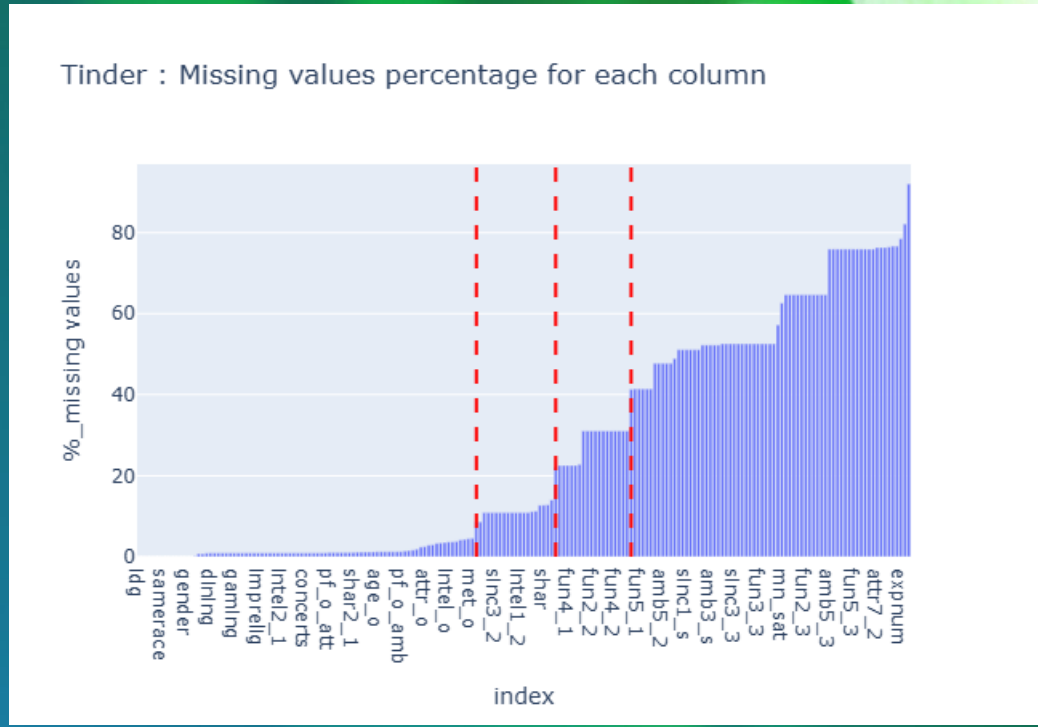
missing values : 26.4%

96 columns usable :

- 13 → 17 colonnes full
- 72 → 69 colonnes avec moins de 5% de valeurs manquantes : imputation possible AVG/median or mode
- Not possible above : imputation too risky for data quality

clean data:

- Data types : int, bool, float, categorial
- Columns rename for better intelligibility
- Replace columns dictionnaries for categorical variables
- Ratings update on same scale : 1-10 -> sum_100 ; handle missing descriptor
- Aggregate data of persons in a DataFrame : **perso_df**
- calculation of additional indicators :
  - success_rate,
  - decision_rate,
  - decision_partner_rate,
  - match_rate... - compute stats per wave : **synth1**



2 EDA of speed daters



Population is mainly between 20 & 30 years, max age woman : 55, man : 42

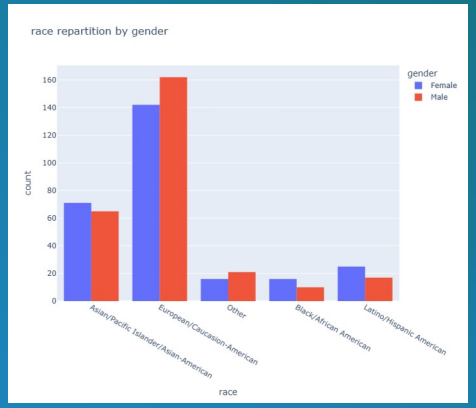
Age repartition : women younger than men,

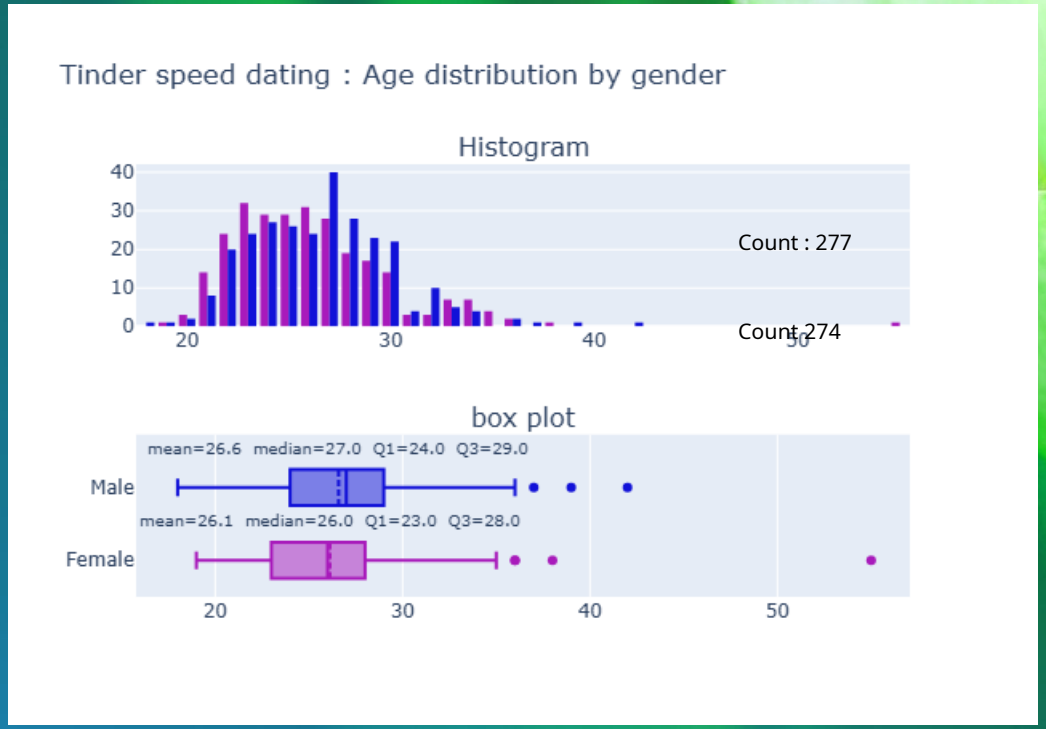
1 year younger in average

Each people is present in one single wave

Race European is over represented

Asian is second,





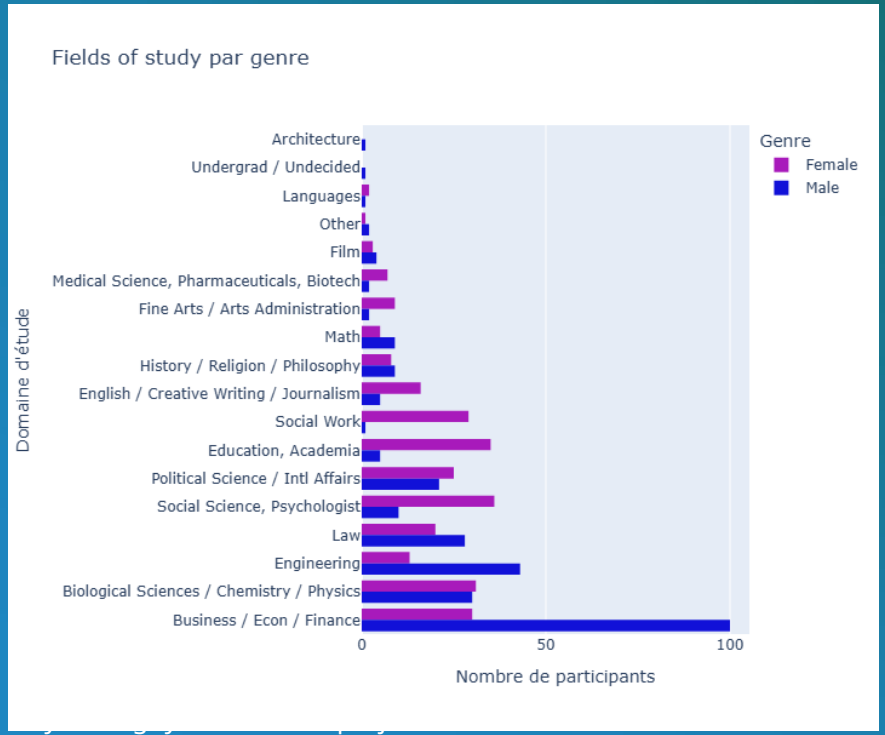
2 EDA of speed daters

Field of study highly genred:

Men : Finance, Engineering, law

Women : Education, social work,

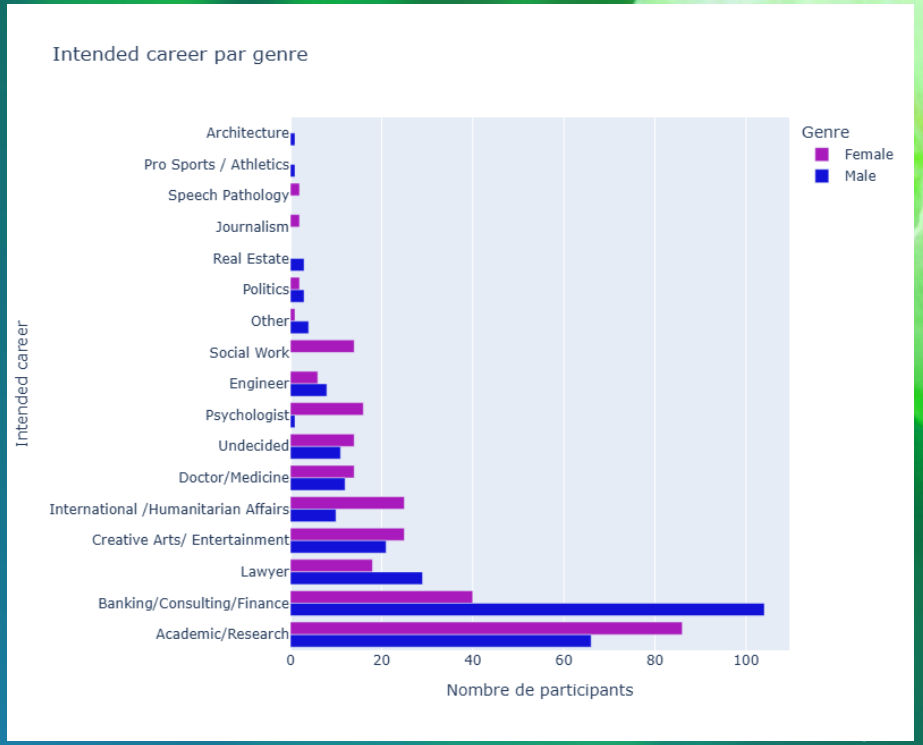
Miss of coherence between field of study and finally intended carreer; A Bias is present as population has a marked mode on 1 or 2 sectors : - A big majority of men intend to work in banking / finance ... or academic research - About Women, they intend to work in Academic research, banking / finance and then huanitarian affairs or creative arts ...



Finaly Intended career :

Miss of coherence with field of study

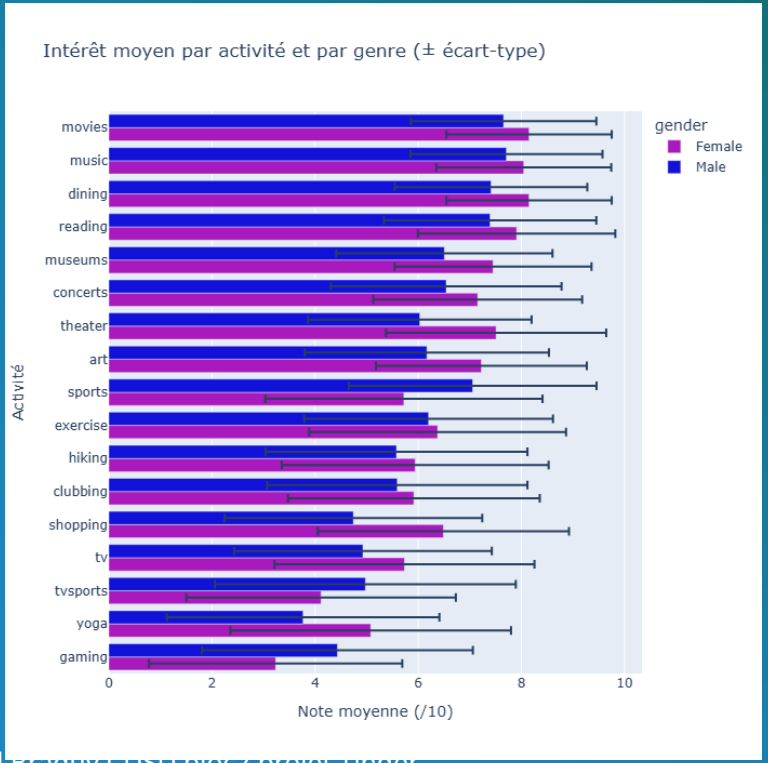
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2 EDA of speed daters

Interests per activity

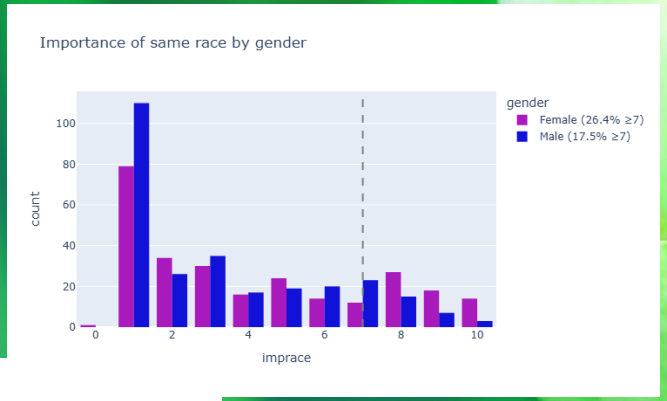
Well balanced between men and women

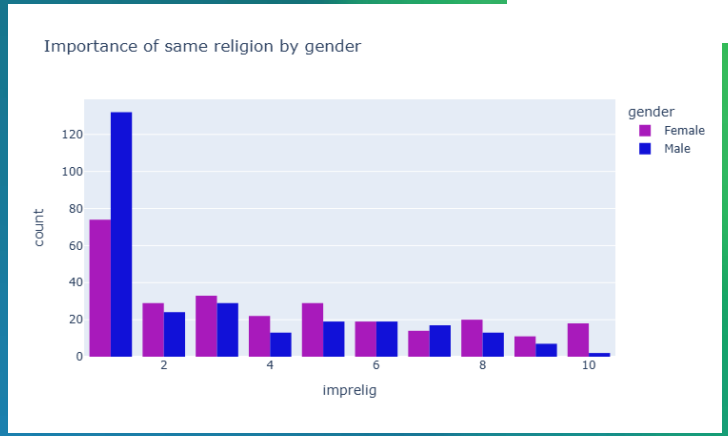


Importance race and religion :

Mainly no importance for men,

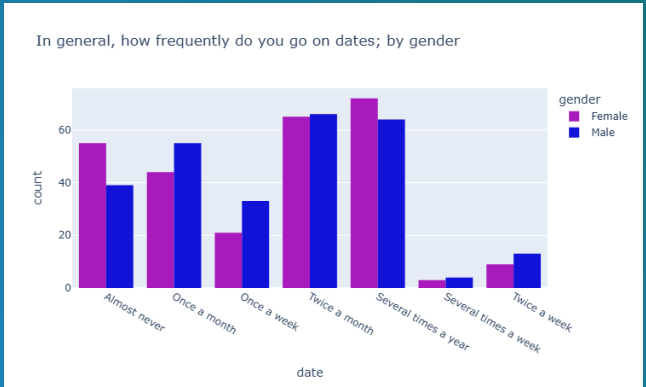
Less For women,



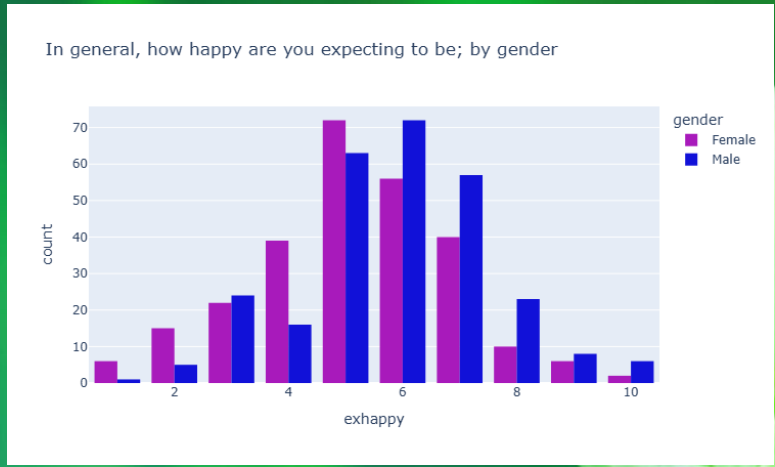


2 EDA of speed daters

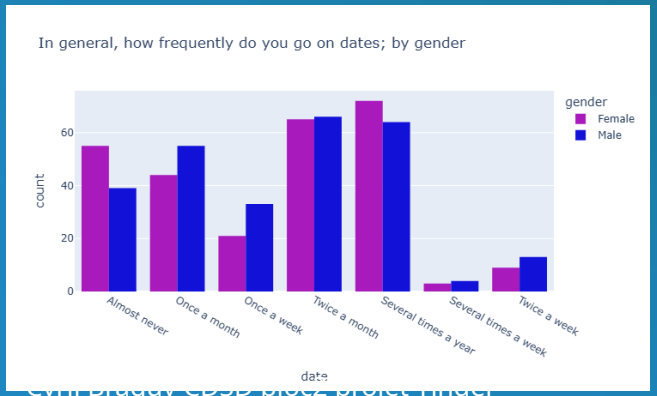
Both go out/ on dates at same frequency

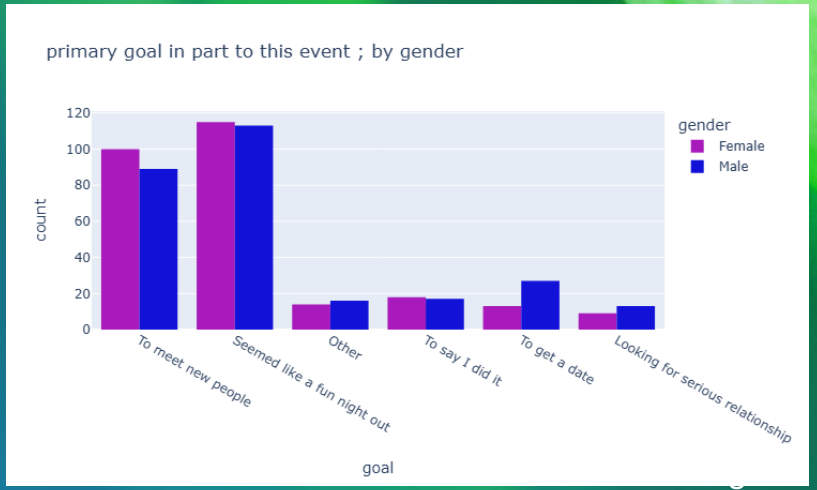


men are more expecting than women



But both are aligned on primary goal

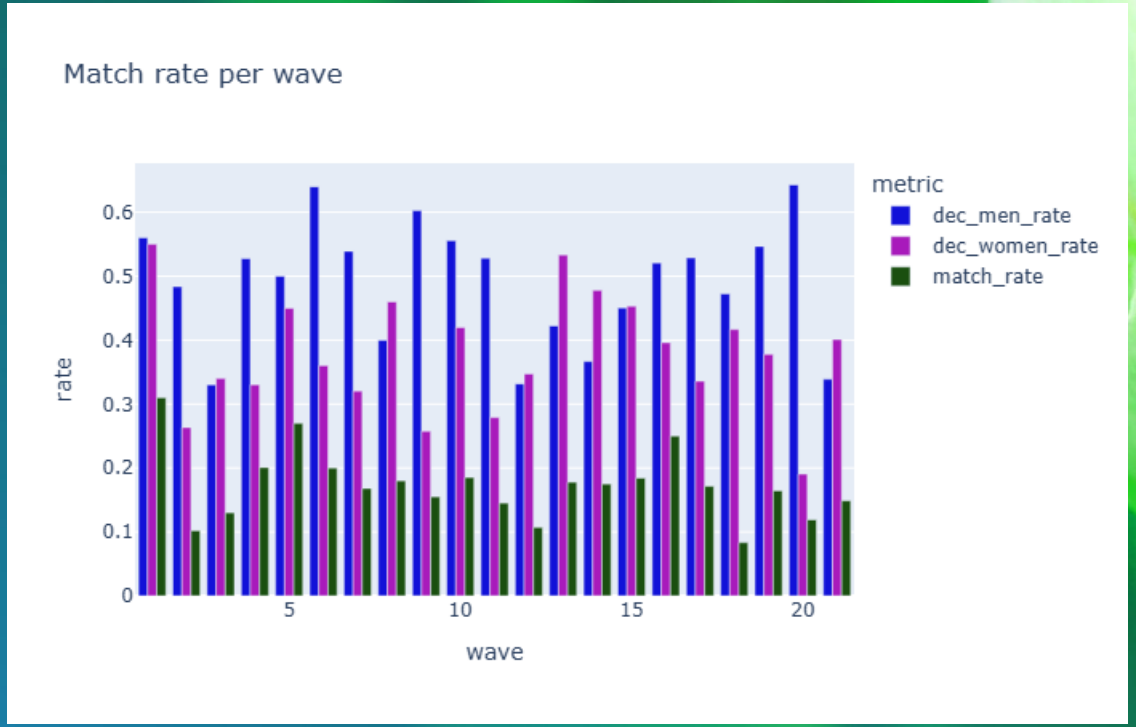




2 EDA of waves

Wave are rather different : match rate is between 10 and 30 %

Men decision id pretty higher than women’s



3 Analysis

Main goan to answer :

What drives people to accept a second date.

One will use intermediate questions :

Q1 : What are the least desirable attributes in a male partner?
Does this differ for female partners?

Q2: How important do people think attractiveness is in potential mate selection vs. its real impact?

Q3: Are shared interests more important than a shared racial background?

Q4 : Can people accurately predict their own perceived value in the dating market?

Q5: In terms of getting a second date, is it better to be someone's first speed date of the night or their last?

3 Analysis

Explo Q1 : What are the least desirable attributes in a male partner? Does this differ for female partners? => dec_o_rate VS

One shall isolate people with **very low rate of partner's decision** and find the trend

According to scatter plots and correlations : Men who have almost no partner's decision are those who are rated by partner :

- low attractivity

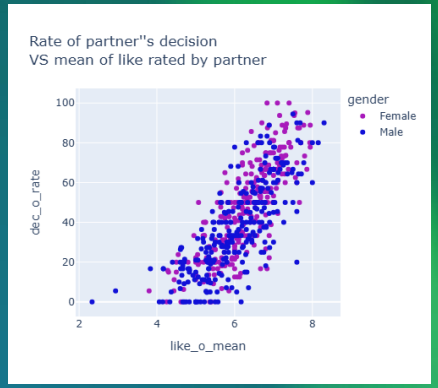
- low fun

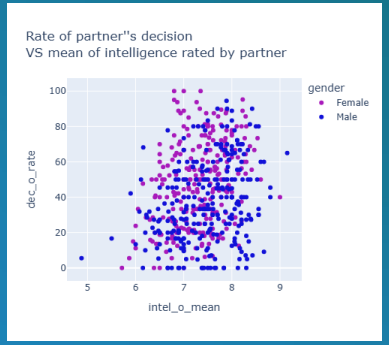
- low shared of interests

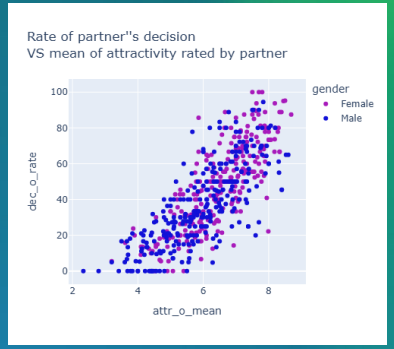
- high intelligence

- high ambition

- high sincerity







correlation between partner's decision and partner's ratings

	Male rated	Female rated
like_o	**0,48**	**0,54**
attr_o	**0,21**	**0,29**
shar_o	**0,20**	**0,18**
fun_o	**0,19**	**0,16**
imprace	**-0,02**	**-0,01**
imprelig	**-0,03**	**-0,03**
age	**-0,03**	**-0,05**
sinc_o	**-0,17**	**-0,25**
amb_o	**-0,21**	**-0,18**
intel_o	**-0,24**	**-0,29**

3 Analysis

Explo Q2: How important do people think attractiveness is in potential mate selection vs. its real impact?

Synthesis :

Men, declare most looking for :

- attractiveness at a big majority
- then intelligence, fun
- and then sincerity

For Women, most important attributes are :

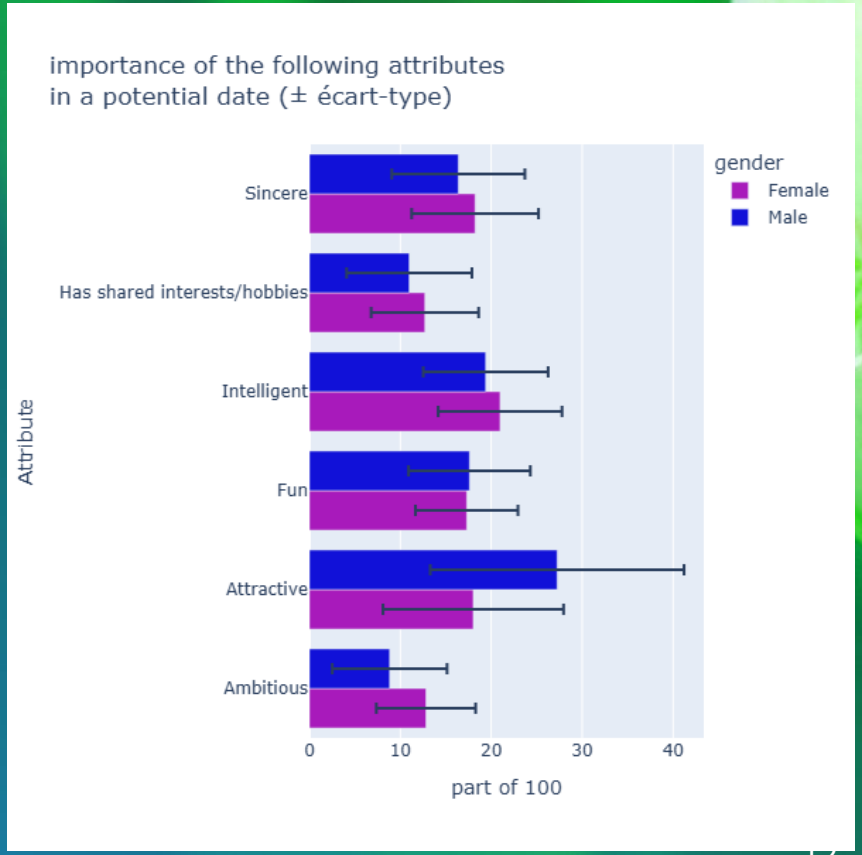
- Intelligence, with a small majority
- then sincerity, fun and attractiveness

These attributes declared by men are opposite to those correlated to their decision's trends : (from more to less important to make a decision Yes :)

Like, attractivenessshared interests, fun

whereas high percieved intelligence ambition and sincerity will cause a no-decision

Same trend for women



3 Analysis

Explo Q3: Are shared interests more important than a shared racial background?

Synthesis :

**Shared interests correlated at resp 20% and 18 %
to decision to have a next date:**

**Shared racial background importance can be a mix of
- imprace : 1-10 scale
- samerace : boolean : if the two dating people are sharng same racial background**

compare importance of num variable (shar_o) and bool variable (samerace) : methods Chi2 and AUC (area under curve)

Samerace has a p-value very low, below 0.05 : the link with decision is close to uncorrelated

Both methods show that shared interests is more correlated than samerace to decision

	Male rated	Female rated
like_o	0,48	0,54
attr_o	0,21	0,29
shar_o	0,20	0,18

Variable	φ / effect	AUC	Conclusion
shar (continous)	—	0.618	correlated
shar_bool (boolean)	0.175	0.589	a bit less correlated, some info lost
samerace	0.023	0.511	almost random link

3 Analysis

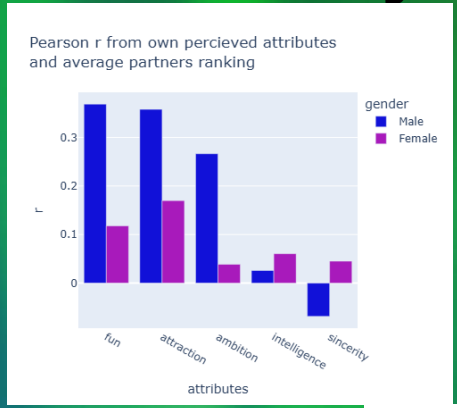
Explo Q4 : Can people accurately predict their own perceived value in the dating market?

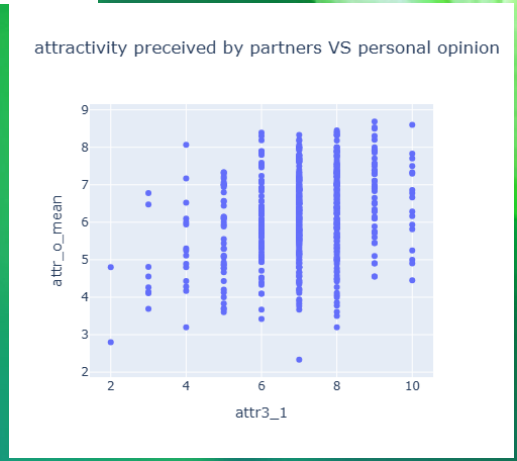
One shall compare people's own vision (attr3_1 and family) and partner's ratngs (attr_o family) mean and std ? correlation ?

For Men, Moderate convergence (r about 0.3) is seen on :
- attractivity
- fun
Low convergence (0.15) is seen on
- ambition

for women, moderate/low convergence seen on attraction, and low on Fun , none on others attributes

But people are not at all able to auto evaluate on intelligence and even less on sincerity





3 Analysis

Explo Q5: In terms of getting a second date, is it better to be someone's first speed date of the night or their last?

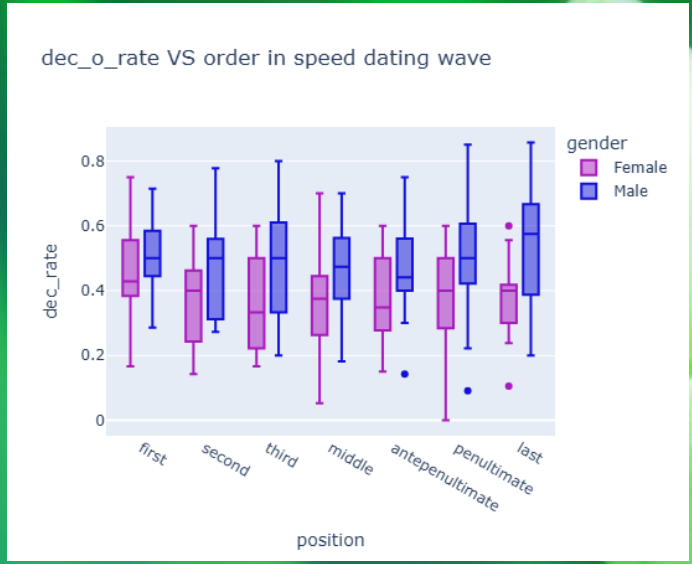
One shall look for correlation between date order of partner (order: integer) // nb of rounds and decision of him.er : dec_o (binary / boolean)

Graphical : no clear correlation is seen

compute biserial correlations between order_rate (numerical variable) and decision (boolean variable) :
correlation is very low, or null r " 0,1 p-val>> 0,05

No clear link can be seen on the global data

Conclusion : no link with order in wave



	order_rate	
	r	P-value
dec	-0,014	0,18
dec_o	-0,013	0,2
match	-0,019	0,08

Global question

What makes people interested into each other to go on a second date together

The global conclusion is that :

average match rate is around 16.5% ; this rate can change to up to 30% depending on wave

Main attributes impacting decision to request a second date are :
a high :
- like
- attraction
- shared interests,
- fun

a low : (for men)
- intell
- ambition
- sincerity

For Women :
- like
- attraction

in addition, almost no impact is seen on:

race, religion, age !! and field of study or career

People are looking for Fun when they date on Tinder !